Jie Lu · Javier Montero · Yi Zhang ·
Junyu Xuan · Tianrui Li · Luis Martínez ·
Etienne Kerre (Eds.)

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Online Fatigue-Aware Power Dispatch for Wind Farms with Hard Feasibility Guarantees

Jiangtao Zhao[1], Lin Zhang[1], Tiantian He[1], Tianyi Zhang[1], Ximeng Wang[2], Bin Wang[3(✉)], and Feng Hong[3]

[1] Sanya Oceanographic Institution, Ocean University of China, Qingdao, China
{zhaojiangtao,zhanglin6624,hetiantian,zhangtianyi9288}@stu.ouc.edu.cn
[2] Cyber Finance Department, Postal Savings Bank of China, Beijing, China
[3] College of Computer Science and Technology, Ocean University of China, Qingdao, China
{wangbin9545,hongfeng}@ouc.edu.cn

Abstract. The rapid expansion of utility-scale wind farms, coupled with high-frequency grid dispatch updates (e.g., 1-second cadence), poses a critical challenge: flexible, large-capacity turbines are increasingly susceptible to amplified cyclic loading and accelerated fatigue accumulation. Traditional uniform power allocation strategies often exacerbate cross-turbine load imbalances, while high-fidelity fatigue assessment methods like Rainflow Counting (RFC) remain computationally prohibitive for real-time control loops. To address this fidelity-latency gap, this study proposes a novel online fatigue-aware dispatch framework that integrates a low-complexity fatigue surrogate with a hard-feasible projection-based scheduler. Unlike existing soft-penalty approaches, our method strictly enforces grid-mandated constraints—including power balance, turbine bounds, and ramp rates—at every time step, guaranteeing operational safety. Crucially, we validate that the proposed lightweight surrogate maintains a high ranking consistency (Spearman correlation > 0.97) with offline RFC benchmarks, enabling effective turbine-level risk prioritization without compromising real-time performance. Experiments on two wind farms demonstrate that the proposed strategy significantly mitigates fatigue damage concentration on critical hotspot turbines (reducing top-K damage variance by up to 48%) while maintaining near-neutral farm-level total damage and strict feasibility. These results establish the framework as a practical and deployable solution for real-time fatigue-aware wind farm control under strict operational constraints.

Keywords: Wind Farm Dispatch · Fatigue-Aware Control · Hard Feasibility · Real-Time Optimization · Load Balancing

J. Zhao and L. Zhang—Co-first authors with equal contributions.

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1 Introduction

Wind power has emerged as a cornerstone of global decarbonization efforts [1], driven by the deployment of increasingly larger and more flexible utility-scale turbines [2]. In modern Automatic Generation Control (AGC) systems, this trend necessitates second-level dispatch updates (typically at a 1-second cadence) that must simultaneously satisfy stringent farm-level power tracking requirements and individual turbine operational limits. While larger rotors enhance energy capture, their inherent flexibility amplifies cyclic structural loading, making fatigue-aware dispatch under hard feasibility constraints a pivotal issue for operational reliability and Levelized Cost of Energy (LCOE) reduction [3].

A fundamental bottleneck in current practice is the mismatch between evaluation fidelity and control latency. Rainflow Counting (RFC), the gold standard for fatigue assessment [4–6], relies on full-cycle identification from historical data, rendering it computationally intractable for online, second-level decision-making [7]. Consequently, real-time dispatch systems often operate without a validated, fatigue-related risk indicator, relying instead on simplified proxies or offline diagnostics that fail to capture dynamic load variations.

At the farm level, prevailing dispatch strategies remain predominantly power-centric [8–10]. Under heterogeneous inflow conditions and complex wake effects, uniform or naive power allocation can induce persistent cross-turbine load imbalances, accelerating fatigue accumulation on specific "hotspot" units [11,12]. While some recent studies have attempted to incorporate fatigue objectives, they largely rely on soft penalty functions within optimization frameworks. Such approaches lack hard feasibility guarantees, risking constraint violations under aggressive wind fluctuations—a scenario unacceptable for grid compliance and equipment safety. Thus, the practical challenge remains: How to enforce strict farm-level balance and turbine-level box/ramp constraints while actively suppressing fatigue concentration in real-time?

To bridge these gaps, we propose an online fatigue-aware dispatch framework tailored for second-level decision-making under hard constraints. Unlike existing fatigue-aware dispatch approaches that rely on soft penalties or offline evaluation, our approach integrates a validated online fatigue ranking signal into a strictly feasible real-time dispatch loop. Our approach combines a computationally efficient fatigue surrogate with a projection-based scheduling algorithm. The technical contributions of this work are threefold:

- We develop a lightweight fatigue surrogate that provides a validated online ranking signal consistent with offline RFC (Spearman > 0.97), enabling real-time turbine-level risk prioritization.
- We design a projection-based scheduler that guarantees strict per-step feasibility (power balance, turbine bounds, and ramp rates), distinguishing the proposed approach from soft-penalty-based methods.
- We integrate online ranking, hard-feasible scheduling, and unified offline RFC evaluation into a coherent real-time dispatch framework, enabling effective fatigue load redistribution under strict operational constraints.

Together, these contributions provide a unified solution for safe, real-time, fatigue-aware dispatch, transforming fatigue management from an offline diagnostic tool into an active, online control layer.

2 Related Work

Power-centric dispatch and control methods substantially improve tracking and grid-level coordination [8,10,12–17], but fatigue is commonly handled as a secondary penalty or an offline diagnostic. As a result, fatigue concentration can persist even when power-level objectives are well satisfied.

System-level reviews summarize broader farm-control advances, including flow control, reactive-power coordination, and offshore scenarios [18–20]. These studies clarify capability landscapes, but they do not directly resolve second-level fatigue-aware dispatch with strict per-step feasibility.

Fatigue modeling studies provide physically grounded cycle-level analysis and life-oriented interpretation [2,21–24]. Their evaluation fidelity is high, yet the resulting pipelines are typically not constructed for low-latency dispatch loops with hard equality/box/ramp constraints.

Data-driven wind-power models improve short-term state observability and can support dispatch decisions [9,25]. However, most are optimized for prediction accuracy rather than for fatigue-aware online redistribution under a strictly feasible dispatch set.

Taken together, the literature still leaves three practical gaps for real-time deployment. First, online fatigue signals are rarely validated for ranking consistency against offline reference damage. Second, fatigue guidance is often detached from a dispatch-feasible second-level control loop. Third, strategy comparisons are frequently reported without a unified post-processing protocol. In particular, existing studies rarely provide an online fatigue signal validated against RFC-based ranking, nor do they integrate such signals into a strictly feasible per-step dispatch framework. The present work addresses these gaps through integrated online ranking, hard-feasible scheduling, and unified offline RFC evaluation.

3 Preliminaries

3.1 Model Assumptions

We use the following assumptions to keep second-level dispatch updates and unified post-processing computationally tractable:

- *Homogeneous turbine type.* Turbines are modeled with identical rated parameters; inter-turbine differences are represented by inflow and command trajectories.
- *Independent per-turbine accounting.* Main-shaft and tower-channel damage indices are accumulated per turbine without explicit cross-turbine structural coupling.

– *Accurate tracking at dispatch timescale.* At the 1 s resolution, issued power commands are treated as accurately tracked; delay-aware closed-loop dynamics are left for future work.
– *Zero-initialized comparative index.* The normalized online index starts from zero for all turbines as a computational baseline for relative comparison.

These assumptions define the modeling scope for second-level dispatch and controlled cross-strategy comparison; absolute structural-life reconstruction remains outside the current scope.

4 Proposed Methodology

4.1 Estimation of Fatigue Damage

This subsection defines the control-oriented online fatigue proxy used in dispatch updates. The proxy is used as a comparative ranking signal across turbines, while final strategy comparison is performed by offline surrogate-based RFC post-processing. We use two load channels throughout: main-shaft torque proxy T_i (shaft channel) and tower-channel load proxy H_i (tower channel). Representative load trajectories for both channels are shown in Fig. 1.

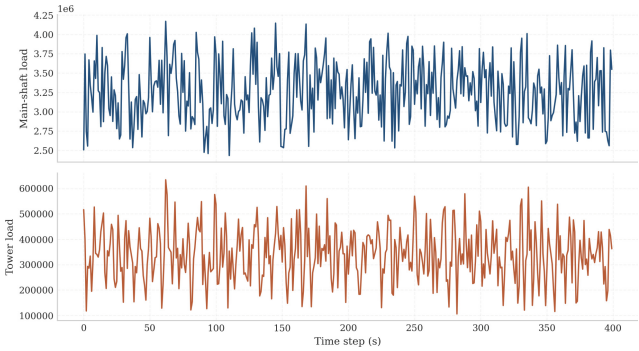

Fig. 1. Representative main-shaft and tower-channel load trajectories for Farm B.

According to the *Palmgren-Miner* linear cumulative damage theory [26], the fatigue damage index D_F can be expressed as:

$$D_F = \frac{n_F}{N_F} \tag{1}$$

where n_F is cycle count, and N_F is fatigue life at that stress level. In online dispatch, L_t denotes the per-channel load proxy at second t, with fixed mapping $L_t^{\text{shaft}} = T_i(t)$ and $L_t^{\text{tower}} = H_i(t)$.

To obtain a low-latency control signal, we treat each second as one proxy increment window and set $n_F = 1$ in Eq. (1). The resulting per-step increment and cumulative proxy are

$$D_{F,t} = \frac{1}{N_{F,t}} \tag{2}$$

$$D_F(T) = \sum_{t=1}^{T} D_{F,t}. \tag{3}$$

The corresponding per-step amplitude and mean stress for channel L_t are

$$S_{a,t} = |L_t - L_{t-1}| \tag{4}$$

$$S_{m,t} = \frac{L_t + L_{t-1}}{2}. \tag{5}$$

Because operating points have nonzero mean stress, we apply Goodman correction [27] to obtain equivalent stress amplitude:

$$S_{\mathrm{eq},i} = \frac{S_{a,i}}{1 - \frac{S_{m,i}}{\sigma_b}} \tag{6}$$

where σ_b is the tensile strength. Using the equivalent stress amplitude $S_{\mathrm{eq},i}$, the fatigue life is estimated via the Basquin S–N relation [28–30]:

$$N_F = \frac{C}{S_{\mathrm{eq},i}^k} \tag{7}$$

with material constants C and k. Accordingly, $N_{F,t}$ follows from Eqs. (6) and (7), and $D_{F,t}$ follows from Eq. (2).

The online proxy is used for comparative ranking in control, and final strategy comparison uses strict offline RFC on reconstructed channels $\hat{T}_i$ and $\hat{H}_i$ under a unified protocol.

Define channel-wise online cumulative proxy as $D^{\mathrm{cum}}_{\mathrm{shaft},i}(t)$ and $D^{\mathrm{cum}}_{\mathrm{tower},i}(t)$, per-step increments as $\Delta D_{\mathrm{shaft},i}(t)$ and $\Delta D_{\mathrm{tower},i}(t)$, and dispatch proxy score as

$$d_i(t) = D^{\mathrm{cum}}_{\mathrm{shaft},i}(t) + D^{\mathrm{cum}}_{\mathrm{tower},i}(t) + \Delta D_{\mathrm{shaft},i}(t) + \Delta D_{\mathrm{tower},i}(t). \tag{8}$$

Here $d_i(t)$ is a dimensionless control index rather than a calibrated physical-life measure: cumulative terms encode long-horizon exposure, while increment terms increase sensitivity to current trend changes. This additive form is a heuristic ranking design for online relative prioritization. Online scheduling uses T_i, H_i, and $d_i(t)$ for relative risk guidance. Offline comparison uses reconstructed $\hat{T}_i$ and $\hat{H}_i$ in strict RFC under the same shaft/tower channel definitions.

4.2 Model of Power Dispatch

The traditional stress-unaware dispatch scheme uses uniform allocation, where the per-turbine power P_{avg} is

$$P_{avg} = \frac{P}{N} \tag{9}$$

where P denotes the total power dispatch instruction issued by the grid to the wind farm at a time step, and N is the total number of turbines in the farm.

However, uniform allocation does not take into account the relationship between turbine operational states and cumulative fatigue damage, resulting in some turbines being subjected to high fatigue loads for extended periods. This could compromise the operational safety of their mechanical components.

To formulate fatigue-aware dispatch as a control problem, command updates must be connected to fatigue-relevant load responses. We therefore introduce control-oriented load proxies before presenting the dispatch objective and constraints.

Control-Oriented Load Proxies. The scheduler acts on power commands, whereas fatigue balancing is guided by predicted load channels. In implementation, both channels are given by fitted surrogate models:

$$T_i(t) = \mathcal{M}_T(\phi_i(t)), \qquad H_i(t) = \mathcal{M}_H(\phi_i(t)), \tag{10}$$

where $\phi_i(t)$ contains $v_i(t), p_i(t), p_i(t-1)$, and optionally includes state-variation features such as pitch, ω_r, ω_f, Δv, and Δp when available. This matches the piecewise-linear load estimator used in the codebase and keeps online dispatch and offline evaluation directionally consistent.

Guiding Dispatch Objective. The dispatch objective is treated as a design principle that balances concentration suppression and command smoothness. At each step, the online fatigue proxy provides balancing guidance, and load-fluctuation regularization controls command variability:

$$J_t \propto \frac{1}{N}\sum_{i=1}^{N}(\Delta D_{\text{shaft},i}(t) + \Delta D_{\text{tower},i}(t)) + \frac{1}{N}\sum_{i=1}^{N}|p_i(t) - p_i(t-1)|. \tag{11}$$

Eq. (11) specifies the intended trade-off between concentration suppression and command smoothness. It is not solved as an independent optimizer; instead, it serves as a control-oriented design principle that guides the construction of the heuristic dispatch updates. Section 4.3 implements this principle via proxy-guided anchoring, smoothness blending, and hard-feasible projection. In implementation, w_f is retained as a fairness-monitoring term in the reported objective proxy, while per-step updates follow the same proxy-guided anchoring, smoothness blending, and hard-feasible projection procedure.

Hard-Feasible Constraint Set. Online dispatch is solved under explicit per-step constraints:

– Farm-level balance:

$$\sum_{i=1}^{N} p_i(t) = P_t \tag{12}$$

– Turbine power bounds:

$$0 \le p_i(t) \le P_{\text{rated}} \quad \forall i \tag{13}$$

– Deviation from farm mean is limited:

$$\left| p_i(t) - \frac{P_t}{N} \right| \le \delta_{\max}, \quad \forall i, \tag{14}$$

– Inter-step ramp is limited:

$$|p_i(t) - p_i(t-1)| \le r_{\max}, \quad \forall i. \tag{15}$$

In the reported setting, $p_{\min} = 0$ MW, $p_{\max} = P_{\text{rated}} = 5$ MW, $\delta_{\max} = 1$ MW, and $r_{\max} = 0.5$ MW/s. For efficient implementation, the three bound types are merged into dynamic per-turbine limits:

$$l_i(t) = \max\left(p_{\min}, \frac{P_t}{N} - \delta_{\max}, p_i(t-1) - r_{\max} \right), \tag{16}$$

$$u_i(t) = \min\left(p_{\max}, \frac{P_t}{N} + \delta_{\max}, p_i(t-1) + r_{\max} \right). \tag{17}$$

If $\sum_i l_i(t) \le P_t \le \sum_i u_i(t)$, the bounded-simplex projection step admits a feasible point and returns a strictly feasible dispatch command.

4.3 Heuristic Dispatch Solver

To realize Eq. (11) under hard constraints, dispatch is computed by a projection-based scheduler with explicit per-step feasibility enforcement.

At each step, let $d_i(t)$ denote the current proxy score defined in Sec. 4.1, combining cumulative terms and the latest increments over shaft and tower channels. The solver first constructs a proxy-guided anchor:

$$\bar{p}_i(t) = \frac{P_t}{N} - \gamma \frac{d_i(t) - \bar{d}(t)}{\sigma_d(t) + \varepsilon}, \tag{18}$$

where $d_i(t)$ is the online proxy score, $\bar{d}(t)$ and $\sigma_d(t)$ are its cross-turbine mean and standard deviation, and $\gamma > 0$ controls balancing strength. To reduce command chattering, the anchor is blended with the previous command:

$$\tilde{p}_i(t) = (1-\eta)\bar{p}_i(t) + \eta p_i(t-1), \quad 0 \le \eta < 1. \tag{19}$$

This yields a smooth candidate command with built-in fatigue balancing.

Power-balance dual updates are then performed under box/deviation/ramp bounds:

$$p_i^{(k)}(t) = \mathrm{clip}\Big(\tilde{p}_i(t) - \lambda^{(k)},\, l_i(t),\, u_i(t)\Big)\,, \tag{20}$$

$$\lambda^{(k+1)} = \lambda^{(k)} + \alpha\left(\sum_{i=1}^{N} p_i^{(k)}(t) - P_t\right), \tag{21}$$

where α is the dual step size. After iterations, a final hard projection onto the bounded simplex is applied:

$$\Pi_{\mathcal{F}_t}(\mathbf{x}) = \arg\min_{\mathbf{p}\in\mathcal{F}_t} \|\mathbf{p}-\mathbf{x}\|_2^2, \quad \mathcal{F}_t = \left\{\mathbf{p} \;\middle|\; \sum_i p_i = P_t,\ l_i \le p_i \le u_i\right\}, \tag{22}$$

which guarantees strict feasibility at each step under the feasibility condition above.

The dual loop serves as a fast pre-balance stage that reduces equality residual and correction magnitude. The final bounded-simplex projection enforces strict hard feasibility.

In implementation, η is the smoothness coefficient from w_s, damage_gain is a base scaling factor, and $\gamma = w_d \times \text{damage_gain}$ is the single effective damage-balancing coefficient used in Eq. (18). Thus, γ is derived from configured weights rather than tuned independently.

4.4 Formulation–Implementation Consistency

Equations (12)–(15) define the implemented feasible set. The dispatch solver is a projection-based heuristic with an explicit fatigue-balancing mechanism, and online dispatch and offline assessment are separated.

For offline assessment, the calibrated surrogate reconstructs loads from wind speed and dispatch commands:

$$\begin{aligned} \hat{T}_i(t) &= \mathcal{M}_T(v_i(t), p_i(t))\,, \\ \hat{H}_i(t) &= \mathcal{M}_H(v_i(t), p_i(t))\,. \end{aligned} \tag{23}$$

In practice, both MT and MH are implemented as lightweight fitted surrogate models calibrated on historical data, ensuring low computational complexity and real-time applicability.

Offline damage is then computed by strict rainflow counting with Goodman correction and Miner accumulation. For strategy s, turbine i, and exponent m_{off}, the damage index is

$$D_{i,s}^{\mathrm{off}}(m_{\mathrm{off}}) = \sum_{c\in\mathcal{R}_{i,s}^{\mathrm{shaft}}} \left(A_{i,s,c}^{\mathrm{shaft}}\right)^{m_{\mathrm{off}}} + \sum_{c\in\mathcal{R}_{i,s}^{\mathrm{tower}}} \left(A_{i,s,c}^{\mathrm{tower}}\right)^{m_{\mathrm{off}}}, \tag{24}$$

with farm-level total $D_s^{\text{off}} = \sum_{i=1}^{N} D_{i,s}^{\text{off}}$. Improvement against baseline b is

$$I_{s\leftarrow b}^{\text{off}}(m_{\text{off}}) = 100 \cdot \frac{D_b^{\text{off}}(m_{\text{off}}) - D_s^{\text{off}}(m_{\text{off}})}{D_b^{\text{off}}(m_{\text{off}})}. \tag{25}$$

All strategy comparisons are performed under this unified protocol. The offline RFC channels are fixed to reconstructed shaft and tower signals $\hat{T}_i$ and $\hat{H}_i$, corresponding one-to-one to the online shaft and tower channels.

5 Experiments

5.1 Data Source

The dataset used in this study contains two real wind-farm sequences, Farm A and Farm B, each with 100 turbines over 2000 s. Per turbine, we use wind speed v, planned or issued power command P_{ref}, generated power P_{out}, main-shaft torque channel T_{shaft}, tower-channel load F_t, and auxiliary states including pitch angle and shaft-speed signals. We evaluate four scenarios: A temporal split, B temporal split, A→B transfer, and B→A transfer. The cross-farm transfer settings (A→B and B→A) are designed to evaluate the generalization ability of the surrogate ranking and dispatch behavior across different wind farm conditions. For each temporal split, the first 1000 steps are used for surrogate fitting and the last 1000 steps are used for online dispatch and offline post-processing. For each transfer setting, fitting is performed on one farm and evaluation uses the first 1000 steps of the other farm. Using the first 1000 steps in transfer avoids mixing source-farm fitting and target-farm temporal-holdout segments.

Table 1 lists the measurement fields, their units, and how each field is used in the dispatch/evaluation pipeline.

Table 1. Data description, units, and channel mapping.

Field	Meaning	Unit	Usage
P_{ref}	planned/issued turbine power command	MW	dispatch input
v	wind speed	m/s	dispatch and surrogate input
P_{out}	measured generated power	MW	tracking/diagnostic check
T_{shaft}	shaft-torque channel	Nm	shaft channel reference
F_t	tower/thrust channel	N	tower channel reference
pitch angle	blade pitch state	deg	auxiliary state
ω_r, ω_f	shaft-speed states	rad/s	torque proxy/auxiliary

All fields are used under a fixed normalization/comparison protocol; no per-scenario preprocessing retuning is applied.

5.2 Experimental Setup

We use a 1-second sampling interval and evaluate all strategies on the same 1000-step test horizon. The compared strategies are:

- *Original farm allocation as the operational reference*: $p_i \propto v_i^3$, followed by hard projection.
- *Uniform comparison baseline*: equal turbine allocation followed by hard projection.
- *Proposed*: damage-guided anchor, dual balance update, and final bounded-simplex projection.

All strategies share the same hard-feasible set defined by Eqs. (12)–(15), and all dispatch/proxy hyperparameters are fixed across scenarios. Notably, these parameters are shared across both temporal and transfer scenarios without per-case retuning, suggesting a degree of cross-scenario robustness.

For evaluation, offline post-processing applies strict rainflow counting with Goodman correction and Miner accumulation under $m_{\text{off}} = 3$. Experiments are run on an Intel i5-9300H CPU with 4 cores, 8 threads, and 16 GB RAM.

Table 2 lists the key symbols, fixed values, and functional roles used in online scheduling and offline post-processing.

Table 2. Key symbols and parameter values used in all scenarios.

Category	Symbol	Value	Role
Power bound	$p_{\min}, p_{\max}$	$0, 5$ MW	constraint
Deviation/ramp	$\delta_{\max}, r_{\max}$	1 MW, 0.5 MW/s	constraint
Dual update	α, K	0.01, 15	solver hyperparameter
Objective weights	w_d, w_f, w_s	$0.6, 0.3, 0.1$	solver hyperparameter
Damage scaling	damage_gain	0.25	base scaling factor
Effective coefficient	$\gamma = w_d \times$ damage_gain	0.15	used in Eq. (18)
Numerical floor	ϵ	10^{-12}	stabilization
S–N scale/exponent	C, k	10^{10}, 3	normalized setting
Online proxy	window, m_{on}	10, 3	online setting
Goodman parameter	σ_b	10^9	normalized constant
Offline RFC	mode, m_{off}	strict, 3	unified protocol

To improve reproducibility, all methods use one protocol: identical scenario splits, feasible-set projection, post-processing pipeline, and fixed hyperparameters with no per-scenario retuning. The uniform strategy is the quantitative baseline for Top-K reduction, while Orig.-prop. is reported as the operational reference.

Runtime and feasibility are reported in Table 3, where the columns separate proposed-only latency statistics from all-strategy feasibility checks. Time is per-step wall-clock latency for the online pipeline, including proxy update, dispatch solve, and final projection, while excluding offline fitting and offline RFC. All methods remain strictly feasible at every step, and the proposed method stays well within the 1-s budget.

Table 3. Runtime and feasibility summary across four scenarios with 1000 steps each.

Metric	Proposed	All strategies
Solve time mean (ms)	8.70–9.28	$\leq$ 9.28
Solve time p99 (ms)	10.60–10.81	$\leq$ 20.43
Solve time max (ms)	13.47–19.37	$\leq$ 27.91
Feasible steps	4000/4000	12000/12000
Balance MAE	$\approx 10^{-14}$, numerical precision level	

5.3 Experimental Results

Results are organized to answer three questions: whether the surrogate is reliable for evaluation, whether the online proxy preserves turbine ranking, and where dispatch gains occur.

We first verify surrogate reliability, because all offline RFC comparisons depend on reconstructed channels.

Table 4. Surrogate fitting quality for two-farm calibration splits.

Fit farm	Split	RMSE_T	R_T^2	RMSE_F	R_F^2
A	Train	85540.52	0.9569	69641.91	0.5937
A	Holdout	87892.00	0.9546	71125.27	0.5878
B	Train	85452.38	0.9566	69382.63	0.5916
B	Holdout	84955.79	0.9572	68603.41	0.6006

In Table 4, the fit quality columns RMSE_T, R_T^2, RMSE_F, and R_F^2 indicate $R_T^2 \approx 0.95$ for both farms with small train–holdout gaps, suggesting limited overfitting. Thrust-channel fitting is lower but stable across splits.

We then test whether the online proxy preserves turbine-level ranking against the offline RFC reference.

Table 5. Online fatigue proxy vs offline RFC damage index consistency for turbine-level ranking under the proposed strategy.

Scenario	Spearman	Kendall	Top-10 overlap
A temporal	0.977	0.884	0.80
A→B	0.971	0.861	0.80
B temporal	0.980	0.885	0.90
B→A	0.953	0.844	0.70

For Table 5, the three ranking columns Spearman, Kendall, and Top-10 overlap jointly indicate high agreement in both temporal and transfer settings; Notably, this consistency remains strong under cross-farm transfer settings, indicating that the online proxy maintains robust ranking performance under dataset shifts. Figure 2 provides the corresponding scatter view.

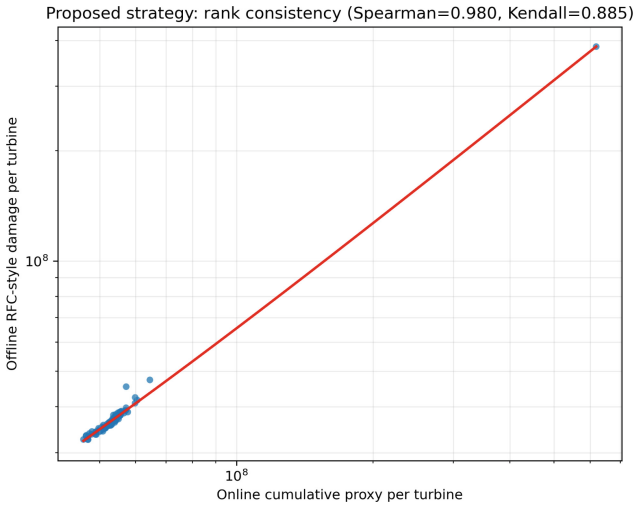

Fig. 2. Turbine-level consistency between online fatigue proxy and offline RFC damage index under the proposed strategy.

We finally evaluate strategy effects under the shared hard-feasible setting.

Across Table 6, the primary improvement is in distributional metrics $D_{\max}$ and $\mathrm{Var}(D_i)$. In the transfer scenarios A→B and B→A, proposed D_{farm} is slightly higher than uniform (4.013 vs 3.989, and 4.114 vs 4.103), while $D_{\max}$ and $\mathrm{Var}(D_i)$ are both reduced. We therefore interpret the method as a redistribution strategy that prioritizes hotspot suppression under hard feasibility, not as a strict minimizer of farm-level total damage.

To interpret where the gains appear in the distribution, Fig. 3 visualizes the B-temporal case. The proposed strategy produces a more compact turbine-level damage distribution and a lighter upper tail than uniform dispatch.

To separate rank-tail behavior from overall distribution shape, Fig. 4 focuses on sorted turbine damage in B-temporal. Gains concentrate in the high-damage tail. Figure 5 then checks whether this tail pattern is stable across scenarios for $K = 1, 3, 5, 10, 20$.

The cross-scenario Top-K curves show monotonic reduction across rank sets, indicating that the effect is not limited to a single extreme turbine.

5.4 Robustness and Mechanism Checks

This subsection reports compact checks on ranking consistency, cross-scenario stability, and bound activity.

Table 7 compares strategy-level rank consistency statistics and shows that the proposed strategy keeps strong proxy–RFC alignment, close to uniform and above the original allocation reference.

Table 6. Scenario-wise comparison with $m_{\text{off}} = 3$ and an evaluation horizon of 1000 steps. Orig.-prop. denotes the original farm allocation reference. Underlined values denote the best result within each scenario group. Units are 10^9 for D_{farm}, 10^8 for $D_{\max}$, and 10^{15} for $\text{Var}(D_i)$.

Scenario	Strategy	D_{farm}	$D_{\max}$	$\text{Var}(D_i)$
A-temp	Orig.-prop.	174.804	26.343	29.647
	Uniform	5.087	5.255	6.878
	Proposed	4.935	4.717	5.254
B-temp	Orig.-prop.	172.789	25.538	14.208
	Uniform	4.039	5.193	2.320
	Proposed	4.000	3.845	1.205
A→B	Orig.-prop.	173.462	26.023	13.850
	Uniform	3.989	5.163	2.296
	Proposed	4.013	4.365	1.593
B→A	Orig.-prop.	173.509	26.659	16.955
	Uniform	4.103	5.200	2.320
	Proposed	4.114	4.329	1.555

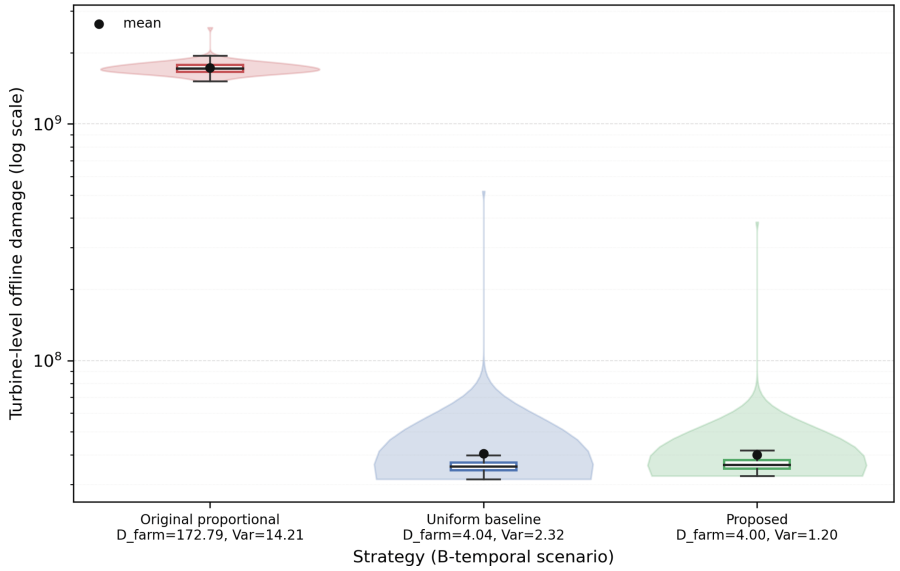

Fig. 3. Distribution of turbine-level offline damage on a log scale in the B-temporal scenario. Violin and box overlays compare the original farm allocation reference, uniform baseline, and proposed strategy.

Table 7. Proxy-vs-RFC ranking consistency across all strategies across four scenarios. Values are reported as mean with minimum-to-maximum range.

Strategy	Spearman	Kendall	Top-10 overlap
Original alloc. (ref.)	0.814 (0.706–0.852)	0.630 (0.522–0.676)	0.58 (0.50–0.70)
Uniform baseline	0.981 (0.978–0.985)	0.887 (0.877–0.905)	0.83 (0.70–0.90)
Proposed	0.970 (0.953–0.980)	0.869 (0.844–0.885)	0.80 (0.70–0.90)

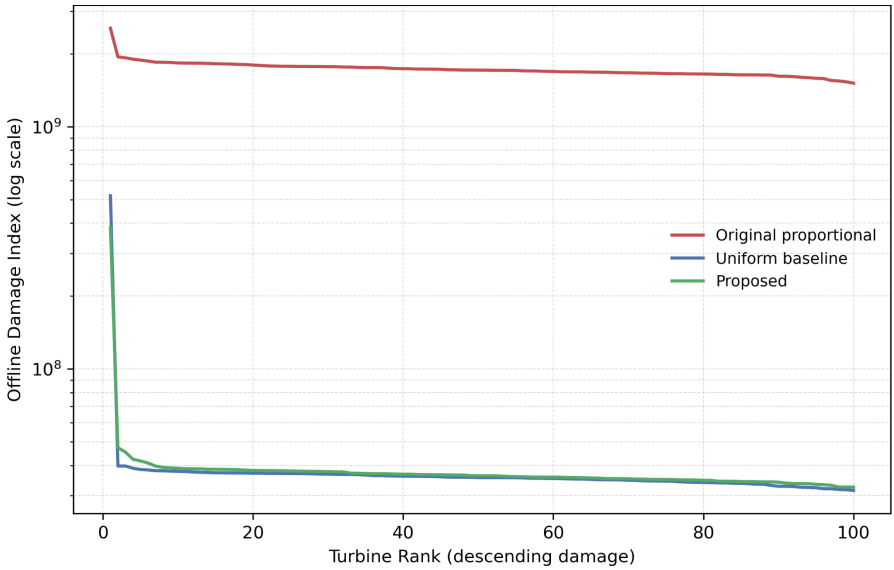

Fig. 4. Sorted turbine offline damage curves in descending order with a log-scale y-axis under three strategies in the B-temporal scenario.

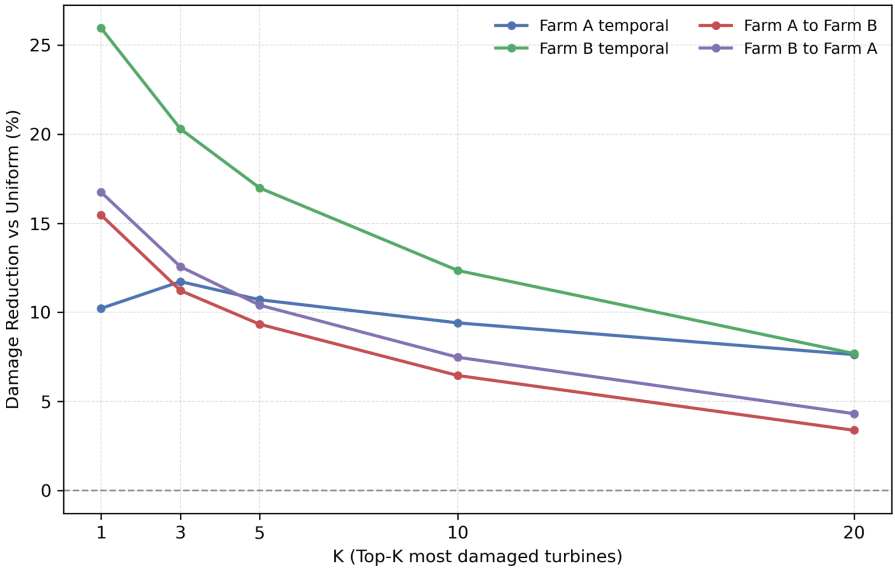

Fig. 5. Cross-scenario Top-K hotspot reduction of proposed strategy vs uniform baseline.

7 Conclusion

This study presents a practical second-level fatigue-aware dispatch framework that combines a lightweight online ranking proxy with hard-feasible projection scheduling. Under a unified offline RFC protocol, the method consistently suppresses high-damage concentration while preserving strict per-step feasibility and near-neutral farm-level total normalized damage. The framework is best interpreted as a comparative control layer for turbine-level fatigue balancing under operational constraints.

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